

Appearance-Based Gaze Estimation with Deep Learning for Interactive Digital Signage

Daniel Miller and Byeong Kil Lee(✉)

Electrical and Computer Engineering, University of Colorado, Colorado Springs (UCCS), Colorado Springs, USA
{dmille15,blee}@uccs.edu

Abstract. The human gaze is a powerful cue for determining the subject of a person's attention. Therefore, the process of estimating a gaze is a crucial component of many human-machine interfaces, providing a more interactive and personal experience. Recent research has demonstrated that deep learning-based approaches to gaze estimation remain highly accurate across a wide range of subjects in unconstrained environments, allowing the technology to be used in a wide range of applications that traditional gaze estimation methods struggle with. One such application for this technology is the growing digital signage industry. This paper proposes the use of real-time deep learning-based gaze estimation in digital signage applications to create a more engaging user experience.

Keywords: deep learning · gaze estimation · digital signage

1 Introduction

Digital signage has become increasingly popular over the years, as traditional methods of public advertising, such as paper posters and billboards, are being phased out in favor of digital signage-based solutions. The digital displays, often in the form of large-format LCD monitors, projectors, and LED video walls, are highly customizable, scalable, and engaging; studies have shown that dynamic digital signage content is far more effective than static traditional signage methods in a retail setting [1]. Furthermore, while the upfront cost of purchasing and installing commercial-grade signage displays is high, the ease of maintaining and managing the digital signage and the ability to remotely update the content as needed often offset the high initial cost.

Existing research shows that digital signage is highly effective in a retail setting, providing a new medium for communicating with customers. In-store marketing has a powerful impact on shopper behavior, leading to a direct increase in shopper engagement. One study shows that the right messaging can drive incremental sales of 5% to 25%, but effective signage is highly dependent on the

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
H. R. Arabnia et al. (Eds.): CSCE 2025, CCIS 2939, pp. 396–410, 2026.
https://doi.org/10.1007/978-3-032-22202-2_30

content of the message, the context of exposure, and the relevance to the audience [2]. When properly applied, the impact of digital signage becomes largely significant.

Digital signage has a strong impact on consumer emotions, which in turn, directly affects buying behavior. Effective use of digital signage can enhance sales and purchases, increase customer engagement, and contribute to an overall positive user experience. A trial conducted by Clear Channel [3] analyzed consumer interaction with a large interactive display placed on a busy London street to determine the impact of adding an interactive element to digital signage. The resulting study analyzed over 2,000 customer interactions and found that 93% of participants agreed that the touch interaction made the advertisement stand out more, and 82% thought the interaction made the advert more engaging [3]. This paper proposes the use of real-time gaze estimation as an alternative to touch-based solutions for user interaction.

Traditional methods of gaze estimation rely on computer vision and fall into three different categories: 2D eye feature regression, 3D eye model recovery, and appearance-based methods. The 2D eye feature regression method learns a mapping function from the geometric eye features to a gaze point, while the 3D eye model recovery method relies on subject-specific 3D eye models to estimate gaze directions. Both methods can deliver accurate results, but they require specialized hardware such as IR cameras and personal calibration for each user. The appearance-based method learns a mapping function from image features to a human gaze, removing the need for specialized hardware, but it only performs well in a constrained environment and is usually subject-specific. The necessity for subject-specific calibration and a constrained environment renders the above methods unsuitable for digital signage applications, which must be able to accurately estimate a gaze from a wide range of users [4]. Recent progress in deep learning based gaze estimation has shown excellent performance improvement in unconstrained environments, opening up technological opportunities for a wide range of novel applications that were previously impractical with traditional gaze estimation methods [5,6]. To address those issues, this paper proposes the use of deep learning gaze estimation to create an interactive digital signage experience for better customer engagement.

The technical contributions presented in this paper include:

- A robust method for mapping pitch and yaw angles from a ResNet model to a specific area of interest on a signage display.
- A process for calibrating the above method to achieve high prediction accuracy without needing to change the training data.
- Measurements of the accuracy prediction and real-time performance of three ResNet architectures across different hardware platforms.

The remainder of this paper is structured as follows. Section 2 discusses the background and motivation, Sect. 3 presents the main ideas. The experimental methodology and analysis are described in Sects. 4 and 5. Section 6 describes the conclusion and future work.

2 Motivation and Problem Statement

Digital signage solutions provide a flexible and scalable advertising platform for retailers and a better customer experience, but typical digital signage installations cannot react to their environment. These displays will continually loop through video content or a slideshow of still images with no opportunity for user interaction, and therefore no ability to customize content for a specific viewer. As an additional consequence, these displays will continue to play the same content even if nobody is present to view it, wasting energy and reducing the display's lifespan. Some signage displays incorporate touch screens to add a layer of interactivity, and while this approach can be highly successful for certain applications, touch displays are not accessible to all users and pose additional requirements such as proximity to the user and frequent cleaning.

Yoshimura et al. proposed the method of dividing the signage display into 12 distinct areas of interest for accuracy measurement [7]. This paper presents a gaze estimation method in which the head pose of the user is estimated using the Seeing Machine Face API to address the impact of head-attitude on existing appearance-based methods. A total of 12 neural networks are used to estimate the gaze direction, one per display area, and each network is trained on 120 image frames.

In this paper, we propose the use of real-time gaze estimation using the L2CS-Net model to provide a far more accessible and engaging digital signage experience [8]. Gaze estimation will be used to estimate a 3D gaze vector, originating from the eye center and pointing toward the gaze subject. By establishing a plane in 3D space to represent the location of the digital display relative to the user, the gaze vector can be used to estimate the specific area of attention on the display. This information can then be used to facilitate user interaction. For example, if a selection of items is presented on a monitor, this method can be used to detect which item the customer is looking at and then provide additional information specific to that item. Furthermore, while this technology provides a hands-free method for the user to interact with content on the display, it is not necessarily constrained to that region. For example, it will take a store fixture with a center-mounted display and a selection of products below the screen. If properly calibrated, the system could detect which product the user is looking at and display the accompanying promotional video on the screen.

3 Main Ideas

3.1 L2CS-Net Architecture

Deep learning-based approaches for gaze estimation offer many advantages over conventional eye model recovery-based and 2D eye feature regression-based methods, leading to widespread popularity. The use of convolutional neural networks (CNNs) allows these methods to effectively extract features from eye images in a non-sensitive way for superior cross-subject gaze estimation performance. In addition, the non-linear nature of a CNN allows it to map eye

appearance to gaze directions across a wide range of head motions, leading to an accurate and robust system [4].

The L2SC-Net model, introduced by Abdelrahman et al., proposes a novel method for estimating 3D gaze angles (yaw, pitch) from an RGB image using a residual network architecture [8]. Unlike other approaches, each gaze angle is regressed independently using two fully connected (FC) layers and two separate loss functions for enhanced angle prediction. These two FC layers share the same convolution layers, using the ResNet50 architecture as the backbone. After the FC layers, a softmax layer with cross-entropy is used to predict binned gaze classifications. Notably, the method proposed by Abdelrahman et al. introduces a novel loss function that linearly combines mean-squared error and cross-entropy loss into the combined loss function to improve network learning. The L2SC-Net model shows remarkable performance in an unconstrained setting and is therefore well-suited for use in a retail setting where a wide range of conditions can be expected.

3.2 Residual Networks

Residual networks are commonly used for image recognition, allowing for deeper networks with better learning capacity while avoiding degradation of training accuracy due to vanishing gradients [9]. This is achieved through the introduction of shortcut connections, which skip the activation of one or more layers to form residual blocks. Residual networks are significantly less complex than comparable VGG networks while offering better performance, requiring less computing power for both training and real-time gaze estimation. It is crucial for digital signage applications, where available computing power is often limited. While the proposed L2CS-Net architecture uses the ResNet50 network as the backbone, this paper will also compare the performance of ResNet34 and ResNet18 networks.

3.3 Data Preprocessing

The Gaze360 dataset proposed by Kellnhofer et al. is used to train the L2SC-Net model [10]. This dataset consists of 238 subjects in both indoor and outdoor environments with labeled 3D gaze estimations in both indoor and outdoor environments, across a wide variety of head poses and distances [10]. The Gaze360 dataset is preprocessed before training; the images are normalized by cropping the eye images from the face images and then applying a rotation and translation to the image to remove the head's roll angle. The annotated 2D gaze vectors are then normalized to the eye center [4].

For real-time gaze estimation, an RGB webcam captures video at a resolution of 640×480. Each video frame must be cropped to a face image. The MediaPipe face-detection solution provided by Google AI implements the lightweight BlazeFace architecture and is used for this purpose due to its fast performance and suitable accuracy for this application [11]. Each video frame is analyzed to determine the bounding box for the face, which is used to crop the image. The cropped

face image is then resized to 448×448 and normalized for input to the network. Figure 1 shows a captured video frame with the face bounding box overlayed, and the resulting cropped face image before resizing and normalization.

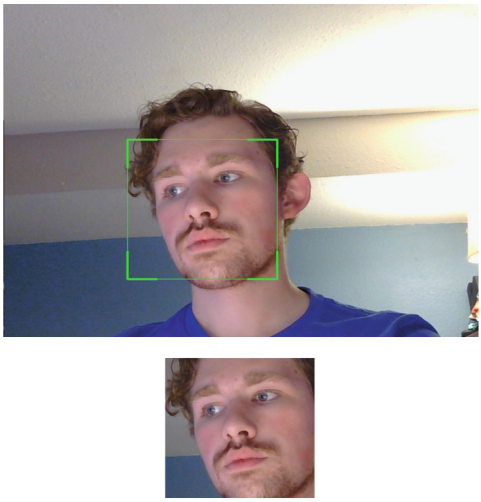


Fig. 1. Video Frame and Cropped Face Image.

3.4 Data Postprocessing

The L2SC-Net model will estimate the gaze direction in the form of yaw and pitch angles. For a digital signage application, these angles will be used to estimate a 2D point of gaze denoting the area of interest on the screen. This paper implements the methodology outlined by Cheng et al. for data postprocessing [4]. First, the yaw and pitch angles are used to calculate the gaze vector in 3D space using trigonometric functions. For this paper, the calculation is performed such that the vector originates from the center of the plot and points along the X-axis. Then, a YZ-plane is established and shifted to a point along the X-axis to represent the screen. The intersection between the 3D gaze vector and the plane produces Y and Z coordinates representing the 2D point of gaze. Figure 2 shows one such gaze vector and the corresponding 2D gaze point on the screen plane. The screen plane has been placed such that it intersects the x-axis at $x = 2$, but this choice is arbitrary. To aid in the visualization, a surface matching the 16×9 aspect ratio of a typical signage display has been shaded in the plot to represent the screen. This 2D gaze point is then mapped to a point on the target display and normalized with respect to the coordinate system of the GUI application.

3.5 Mapping Gaze Point to Pixel Coordinate

The 2D gaze point, which is determined from the intersection of the 3D gaze vector and the screen plane, is mapped to a pixel coordinate on the digital signage display. This mapping depends on many factors, including the physical size and resolution of the display, the location of the camera, and the relative location of the user; as a result, a simple linear relationship is often not sufficient to map a point on the target display with acceptable accuracy.

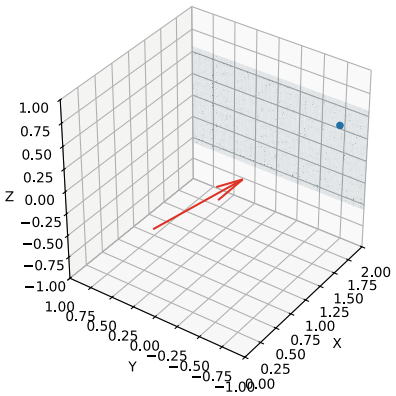


Fig. 2. 3D Gaze Vector and Point of Intersection.

In this work, a homograph matrix is used to describe the translation between the plane on which the 2D gaze point lies and another plane representing the physical screen area of the signage display, allowing a gaze point to be accurately mapped to a pixel coordinate, accounting for shifts in perspective. OpenCV is used to compute the homograph matrix from a set of four source points and a set of four destination points to find the most probable perspective transformation. The four destination points for this process are placed near the four corners of the signage display, slightly spaced from the edges. The specific signage display used for this work has a screen resolution of 1920×1080, and the destination points are placed 150 pixels out from the edges of the display. The exact pixel coordinates of each of the four destination points are listed in Table 1. Centered around these points are green squares with a side length of 100 pixels, serving as the visual target for the user to look at during the computation of the homograph matrix. The calibration screen is shown in Fig. 3. The source points must be determined through a calibration process, discussed in the next section.

Table 1. Exact Pixel Coordinates of Destination Points

Corner	Pixel Coordinate (X, Y)
Top Left	(150, 930)
Top Right	(1770, 930)
Bottom Left	(150, 150)
Bottom Right	(1770, 150)

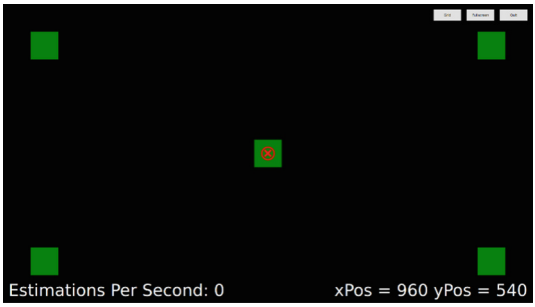


Fig. 3. Calibration Screen with Destination Targets.

3.6 Measuring Source Points for Perspective Mapping

The four source points for the computation of the homograph matrix are determined through a manual calibration process. To begin, the user should stand a set distance away from the display, with their face aligned to the center of the

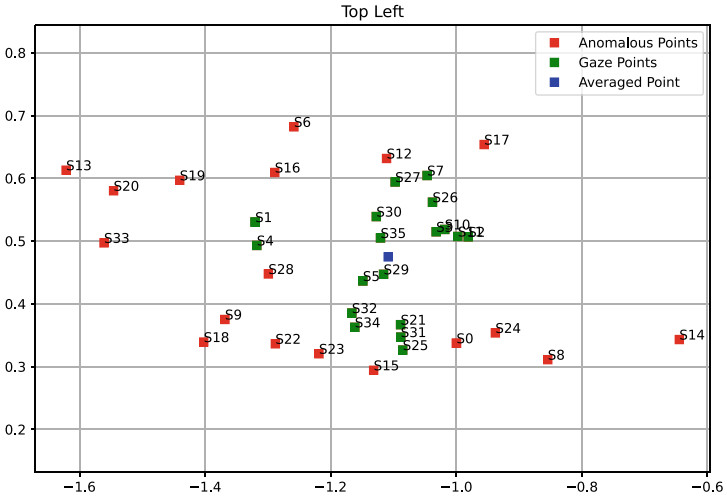


Fig. 4. Collected 2D Gaze Points for Upper Left Corner.

screen. This distance should remain consistent with the expected usage distance after calibration. The user will then look at each of the four corner targets shown in Fig. 3. For each target, three seconds of continuous gaze data are collected to

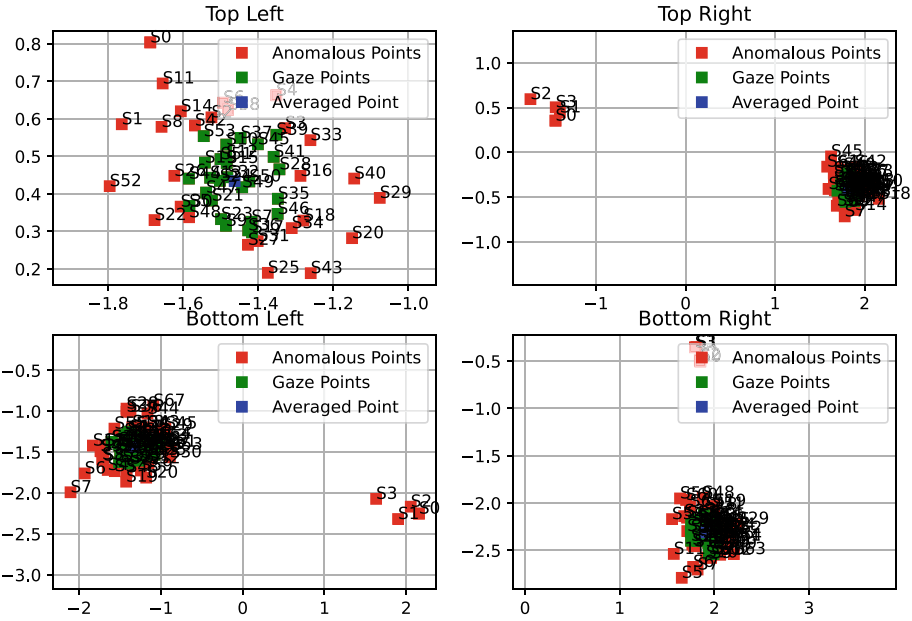


Fig. 5. 2D Gaze Points for All Corners.

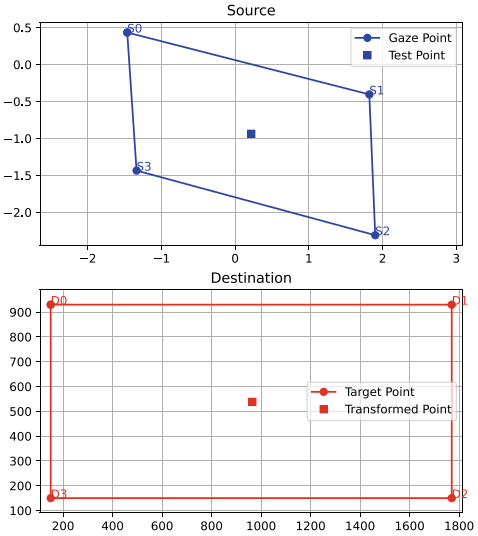


Fig. 6. Homograph Transformation.

correlate the gaze point with the physical location of the target on the display. The total number of collected gaze points depends on the real-time performance of the L2SC-Net architecture on the target hardware; in this work, roughly 60 datapoints were collected for each corner.

Due to the significant level of variation in real-time gaze data, these samples must be filtered to remove outliers. The data does not follow a normal distribution, so the isolation forest algorithm is used to isolate and remove anomalous datapoints from the dataset with a contamination level of 50% [12]. After filtering, the remaining data points are averaged to determine the estimated gaze point for the target. Figure 4 shows a scatter plot of the collected datapoints for the upper left corner target, in which the isolated datapoints are marked red and the estimated gaze point is marked blue. Figure 5 shows gaze data for all four corners.

These gaze points are then used as the four source points for the computation of a perspective transformation between the two planes. Figure 6 shows the results after a calibration; the top plot shows the four estimated gaze points mapping the boundaries of the target display, and for demonstration purposes, a point is placed midway between the lower left and upper right corners of the display. The bottom plot shows the destination coordinates at the center of each target on the display. The midpoint between the gaze points is mapped to a pixel coordinate on the display.

4 Experimental Methodology

4.1 Network Training

Pretrained weights for the ResNet18, ResNet34, and ResNet50 networks are used for gaze estimation. These weights were trained on the normalized Gaze360 dataset using a PyTorch-based implementation of the L2SC-Net architecture. The Adam optimizer was applied with a learning rate of 0.00001, and all three networks were trained over the span of 100 epochs with a batch size of 64.

The Gaze Angular Error is used as the evaluation metric for the trained networks. Assuming that the actual gaze direction is $\mathbf{g} \in \mathbb{R}^3$ and the estimated direction is $\hat{\mathbf{g}} \in \mathbb{R}^3$, the angular error can be computed as:

$$L_{\text{angular}} = \frac{\mathbf{g} \cdot \hat{\mathbf{g}}}{\|\mathbf{g}\| \, \|\hat{\mathbf{g}}\|}$$

4.2 Experiment Setup

The video is captured from a Logitech Brio 100 webcam at a resolution of 640×480. This camera is placed on top of the signage display, horizontally aligned with the center of the screen. All tests are performed using a $50''$ Samsung DC49H commercial monitor with a resolution of 1920×1080. For calibration and testing, the user will view the display from a distance of four feet, their face aligned with the display center.

Real-time estimation is performed using a PyTorch-based gaze-estimation platform based on the L2SC-Net architecture. The estimated pitch and yaw angles are used to calculate a 2D point of gaze, which is then mapped to a pixel coordinate on the target display. This paper adopts a similar classification system employed by Yoshimura et al. where the signage display is divided into a grid of 12 areas of interest (4 horizontal and 3 vertical) [7]. To visualize this in real time, a PyQt5-based UI is developed that divides the target display into a grid of boxes as shown in Fig. 7. The pixel coordinate of the mapped gaze target is represented by a red target, and the box over which this target lies is lit up to visually represent the area of interest. It is not expected that a digital signage display will need a high degree of precision, and the degree of granularity in the subdivisions can be freely adjusted.

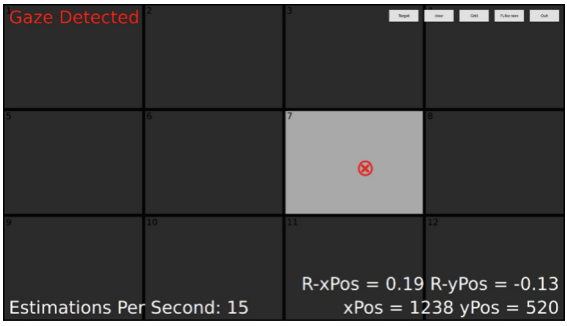


Fig. 7. Display Areas of Interest and Gaze Point.

4.3 Accuracy Measurements

To measure the estimation accuracy, a system is employed in which the user is prompted to look at a green target at the center of the areas, one by one. Similar to the calibration process, three seconds of continuous gaze data is collected at for each area. If the average 2D point of gaze from this data falls within the targeted area, the estimation is considered accurate for that area. The final accuracy of each model will be determined by the percentage of accurately predicted areas. To help visualize the process later in the paper, areas in which the gaze is determined to be accurate are marked green, and the areas considered inaccurate are marked red.

5 Results and Analysis

5.1 Estimated Gaze Point Accuracy

The PyTorch implementation of the L2CS-Net architecture is used with the pre-trained ResNet18, ResNet34, and ResNet50 weights to estimate the 2D point of

gaze for each area on the display. If the average gaze point over three seconds falls within the target area, it will be marked green, as explained in the experiment setup. All tests are performed using a 50″ display, and the user is 4 ft from the screen center. First, ResNet50 is tested with the display divided into 12 distinct areas (4 horizontal and 3 vertical). This test is repeated three times, and the resulting accuracy maps are shown in Fig. 8. The performance across all three tests was consistent, and average accuracy for ResNet50 is 94.4.

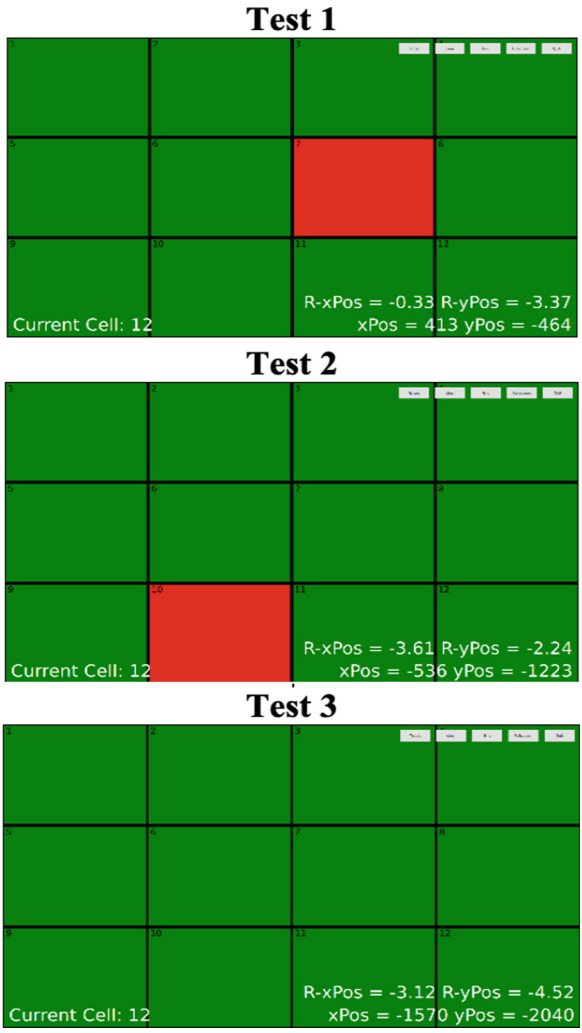


Fig. 8. Accuracy Maps for ResNet50.

This process is repeated for ResNet34. The accuracy suffers near the display center across all three tests. The first accuracy map is shown in Fig. 9, and the average accuracy is 72.22.



Fig. 9. Accuracy Map for ResNet34.

The process is repeated a third time for ResNet18, and the results for the first accuracy map are shown in Fig. 10. It appears that during the calibration for the second test, the estimated gaze point for the upper left target was significantly misaligned, skewing the estimations in that area. The performance with the ResNet18 model was quite poor, with an average accuracy of 61.11.

As a final test, the ResNet50 model is used once again, however, the granularity of the subdivisions is doubled for 48 distinct areas (8 horizontal and 6 vertical). Two tests are performed, and the results are far less consistent (Fig. 11). The accuracy of the first test was 29.16%, and the second test was twice as accurate at 60.14%. This demonstrates the importance of accurate calibration, as a misaligned gaze point at one corner will greatly impact the computed homograph matrix and the resulting translation.

Figure 12 compares the performance of the three networks across the three tests performed. As a general rule, the accuracy of the estimations is directly proportional to the complexity of the model. ResNet18 provides inconsistent performance and would be unsuitable for a digital signage application. ResNet34 performed marginally better, but ResNet50 should be prioritized despite its higher complexity, resulting in worse real-time performance.

5.2 Real-Time Performance

Adequate real-time estimation performance is crucial for a responsive user experience. The PyTorch implementation of the L2SC-Net architecture discussed in this paper benefits significantly from the use of an Nvidia GPU with available CUDA cores, and reasonable performance was not achieved with a CPU. A digital signage display is not expected to have significant computing power available and power efficiency is a priority, so the benchmark is performed on the following devices: a low-power Nvidia T550 GPU intended for use in laptop computers, an Intel i7-1255U to measure performance on a CPU, and an Apple M4 SOC to

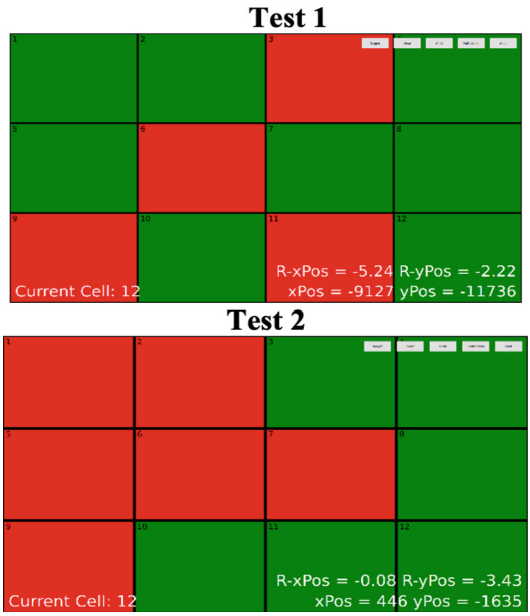


Fig. 10. Accuracy Map for ResNet18.

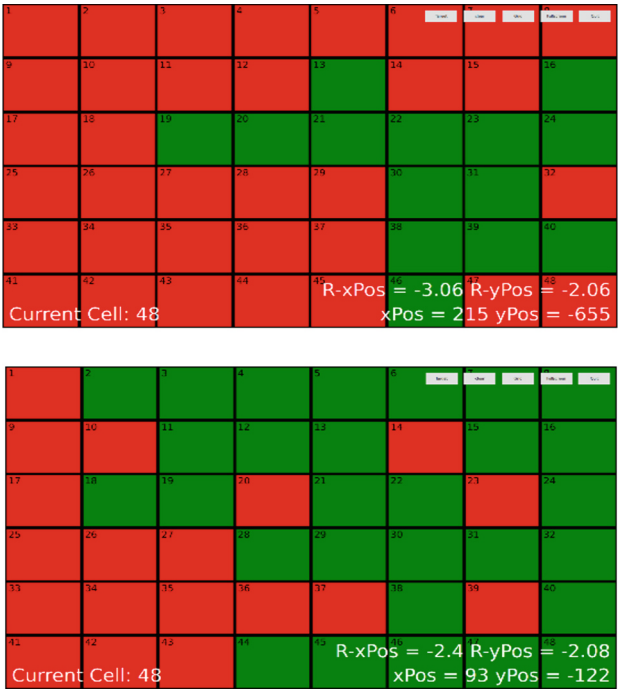


Fig. 11. Accuracy Map for ResNet50 with 48 Areas.

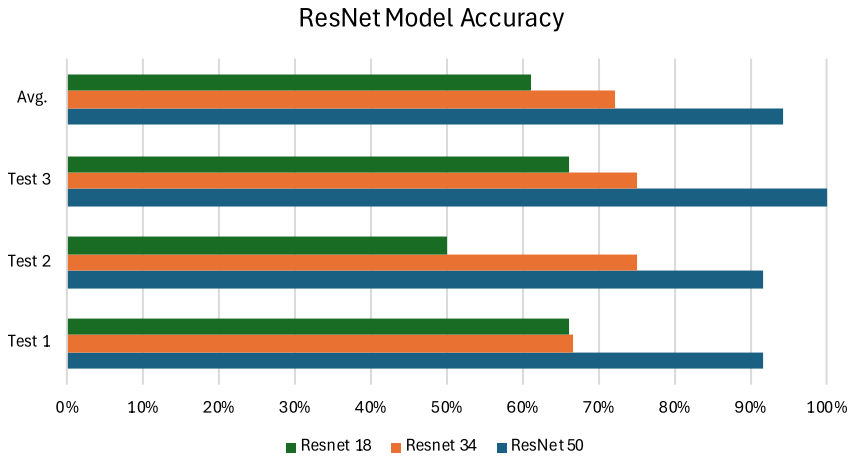


Fig. 12. Accuracy Map for ResNet 50 with 48 Areas.

measure performance using the Metal API. The results in Table 2 demonstrate usable performance on all platforms, but the Cuda and Metal backends provided by the Nvidia GPU and Apple SOC show that hardware acceleration, even on low-power devices, should be used for an optimal experience.

Table 2. Estimations Per Second Benchmark

Device	Backend	Architecture	Estimations Per Second
Nvidia T550	CUDA	ResNet18	27.6
		ResNet34	26.4
		ResNet50	21.0
Intel i7-1255U	CPU	ResNet18	5.8
		ResNet34	3.6
		ResNet50	2.6
Apple M4	MPS	ResNet18	19.76
		ResNet34	19.775
		ResNet50	25.3

6 Conclusion and Future Work

Deep learning-based gaze estimation shows remarkable potential for interactive digital signage applications. This work shows that real-time gaze estimation with the L2SC-Net architecture can be used to estimate the point of gaze on a digital display allowing for user engagement. The method proposed for computing the

2D gaze point from the 3D gaze vector and mapping that gaze point to a pixel coordinate on the target display shows promising performance with the ResNet50 architecture. On a 50-in. display divided into 12 distinct areas, the predictions remain accurate across the entire surface of the display with an average accuracy of 94%. However, when the display is further subdivided into 48 distinct areas, the accuracy drops significantly to 44.79%.

Additional information, such as whether a gaze is detected, can be used to put the display into a lower power state if there is nobody around, or enter an attract mode to get the attention of nearby users. Further improvements in accuracy and better use of the gaze target information can help solidify real-time gaze estimation as the next step forward in digital signage technology.

References

1. Dennis, C., Brakus, J., Gupta, S., Alamanos, E.: The effect of digital signage on shoppers' behavior: the role of the evoked experience. J. Bus. Res. (2014)
2. Burke, R.: Behavioral effects of digital signage. J. Advert. Res. JAR **49** (2009)
3. McMurray, I.: Trial demonstrates effectiveness of interactive digital signage installation-international (2012)
4. Cheng, Y., Wang, H., Bao, Y., Lu, I.: Appearance-based Gaze Estimation with Deep Learning: A Review and Benchmark. State Key Laboratory of Virtual Reality Technology and Systems, SCSE, Beihang University, China (2014)
5. Zhang, X., Sugano, Y., Fritz, M., Bulling, A.: It's written all over your face: full-face appearance-based gaze estimation. In: The IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 2299–2308 (2017)
6. Valikhujaev, Y.: MobileGaze: pre-trained mobile nets for gaze-estimation. Zenodo. https://doi.org/10.5281/zenodo.14257640. Accessed 24 Oct 2024
7. Yoshimura, H., Hori, M., Shimizu, T., Iwai, Yo.: Appearance-based gaze estimation for digital signage considering head pose. Int. J. Machine Learn. Comput. **5**, 507–511 (2015)
8. Abdelrahman, A., Hempel, T., Khalifa, A., Al-Hamadi, A.: L2CS-Net: fine-grained gaze estimation in unconstrained environments. In: 8th International Conference on Frontiers of Signal Processing (ICFSP), pp. 98–102 (2023)
9. He, K., Zhang, X., Ren, S., Sun, J.: Deep Resudual Learning for Image Recognition. Microsoft Research (2015)
10. Kellnhofer, P., Recasens, A., Stent, A., Matusik, W., Torralba, A.: Gaze360: physically unconstrained gaze estimation in the wild. In: International Conference on Computer Vision (ICCV) (2019)
11. Bazarevsky, V., Kartynnik, Y., Vakunov, A., Raveendran, K., Grundmann, M.: BlazeFace: Sub-millisecond Neural Face Detection on Mobile GPUs (2019)
12. Liu, F.T., Ting, K., Zhou, Z.-H.: Isolation Forest, pp. 413–422 (2009)